1. NAME OF THE MEDICINAL PRODUCT

Allerzal F.C. Tablet 5mg

2. PHARMACEUTICAL FORM AND STRENGTH

Film-coated tablet: White to off-white, oblong and biconvex tablet; engraved with "293" on one surface and "STD" on the other surface, containing 5mg of Levocetirizine Dihydrochloride.

List of excipients include: microcrystalline cellulose, lactose (spray-dried), magnesium stearate, hydroxypropylmethyl cellulose, polyethylene glycol 400, and titanium dioxide.

3. THERAPEUTIC INDICATION

Levocetirizine is indicted for the symptomatic treatment of allergic rhinitis (including persistent allergic rhinitis) and chronic idiopathic urticaria.

4. POSOLOGY AND METHOD OF ADMINISTRATION

It must be taken orally, swallowed whole with liquid and may be taken with or without food.

Adults and adolescents 12 years and above:
The recommended dose is 5mg once daily (1 film-coated tablet).

Children aged 6 to 12 years:
The recommended dose is 5mg once daily (1 film-coated tablet).

Elderly:
Adjustment of the dose is recommended in elderly patients with moderate to severe renal impairment (see Patients with renal impairment below).

Adult patients with renal impairment:
The dosing intervals must be individualized according to renal function. Refer to the following table and adjust the dose as indicated. To use this dosing table, an estimate of the patient's creatinine clearance (CL_{cr}) in ml/min is needed. The CL_{cr} (ml/min) may be estimated from serum creatinine (mg/dl) determination using the following formula:

$$CL_{cr} = \frac{[140 - \text{age (years)}] \times \text{weight (kg)}}{72 \times \text{serum creatinine (mg/dl)}} \quad (\times 0.85 \text{ for women})$$

Dosing Adjustments for Patients with Impaired Renal Function:

Group	Creatinine clearance (ml/min)	Dosage and frequency
Normal	≥ 80	1 tablet once daily
Mild	50 – 79	1 tablet once daily
Moderate	30 – 49	1 tablet once every 2 days
Severe	< 30	1 tablet once every 3 days
End-stage renal disease - Patients undergoing dialysis	< 10	Contra-indicated

In paediatric patients suffering from renal impairment, the dose will have to be adjusted on an individual basis taking into account the renal clearance of the patient and his body weight.

Patients with hepatic impairment:
No dose adjustment is needed in patients with solely hepatic impairment. In patients with hepatic impairment and renal impairment, adjustment of the dose is recommended (see Patients with renal impairment above).

5. CONTRAINDICATIONS

Hypersensitivity to levocetirizine, any piperazine derivatives or any constituent of the formulation.

Patients with severe renal impairment at less than 10 ml/min creatinine clearance and patients undergoing haemodialysis.

Patients with rare hereditary problems of galactose intolerance, Lapp lactase deficiency or glucose-galactose malabsorption.

6. WARNINGS AND PRECAUTIONS

Precaution is recommended with intake of alcohol or other central nervous system (CNS) depressants (see Interactions).

Effects on the ability to drive or operate machinery: Comparative clinical trials have revealed no evidence that levocetrizine at the recommended dose impairs mental alertness, reactivity or the ability to drive. Nevertheless, some patients could experience sommolence, fatigue and asthenia under therapy with levocetrizine. Therefore, patients intending to drive, engage in potentially hazardous activities or operate machinery should take their response to the medicinal product into account.

Caution should be taken in patients with predisposing factors of urinary retention (e.g. spinal cord lesion, prostatic hyperplasia) as levocetrizine may increase the risk of urinary retention.

7. INTERACTION WITH OTHER MEDICINAL PRODUCTS AND OTHER FORMS OF INTERACTION

No information on interaction has been reported with levocetirizine (including CYP3A4 inducers); information reported with the racemate compound cetirizine demonstrated that there were no clinically relevant adverse interactions (with antipyrine, pseudoephedrine, cimetidine, ketoconazole, erythromycin, azithromycin, glipizide and diazepam). A small decrease in the clearance of cetirizine has been reported with theophylline; while no alteration in the disposition of theophylline. It is possible that higher theophylline doses could have a greater effect.

Increased exposure to cetirizine and slight altered disposition of ritonavir have been reported with concomitant use of ritonavir and cetirizine.

The extent of absorption of levocetirizine is not reduced with food, although the rate of absorption is decreased.

In sensitive patients, the simultaneous administration of cetirizine or levocetirizine and alcohol or other CNS depressants may have effects on the central nervous system, although it has been reported that the racemate cetirizine does not potentiate the effect of alcohol.

8. PREGNANCY AND LACTATION

Fertility
There are no relevant data available.

Pregnancy
Caution should be exercised when prescribing to pregnant women. For levocetirizine no clinical data on exposed pregnancies are available. Animal studies do not indicate direct or indirect harmful effects with respect to pregnancy, embryonal/foetal development, parturition or postnatal development.

Lactation
Caution should be exercised when prescribing to lactating women. Cetirizine is excreted in human breast milk. Levocetirizine is also expected to be excreted in human milk.

9. EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

In clinical trials the occurrence of somnolence, fatigue, and asthenia has been reported in some patients under therapy with levocetirizine. Patients should be cautioned against engaging in hazardous occupations requiring complete mental alertness, and motor coordination such as operating machinery or driving a motor vehicle after ingestion of Levozine.

10. ADVERSE REACTIONS

Adverse reactions are ranked under headings of frequency using the following convention:

Very common > or equal to 1/10
Common > or equal to 1/100 to < 1/10
Uncommon > or equal to 1/1000 to < 1/100
Rare > or equal to 1/10000 to < 1/1000
Very rare < 1/10000
Not known (cannot be estimated from the available data)

Nervous system disorders
Common: headache, somnolence

Gastrointestinal disorders
Common: dry mouth
Uncommon: abdominal pain

General disorders and administration site conditions
Common: fatigue
Uncommon: asthenia

Post-marketing Data

Very rare cases of the following adverse drug reactions have been reported in post-marketing experience:

Immune system disorders
Not known: hypersensitivity including anaphylaxis

Metabolism and nutrition disorders
Not known: increased weight

Psychiatric disorders
Not known: aggression, agitation, hallucination, depression

Nervous system disorders
Not known: convulsions, paraesthesia, dizziness

Eye disorders
Not known: visual disturbances, blurred vision

Cardiac disorders
Not known: palpitations, tachycardia

Respiratory, thoracic and mediastinal disorders
Not known: dyspnoea

Gastrointestinal disorders
Not known: nausea, vomiting

Hepatobiliary disorders
Not known: hepatitis, abnormal liver function test

Skin and subcutaneous tissue disorders
Not known: angioneurotic oedema, fixed drug eruption, pruritus, rash, urticaria

Musculoskeletal and connective tissue disorders
Not known: myalgia

Renal and urinary disorders
Not known: dysuria

11. OVERDOSAGE

Overdosage has been reported with levocetirizine. Symptoms of overdose may include drowsiness in adults and initially agitation and restlessness, followed by drowsiness in children. There is no known specific antidote to levocetirizine. Should overdose occur, symptomatic or supportive treatment is recommended. Levocetirizine is not effectively removed by dialysis, and dialysis will be ineffective unless a dialyzable agent has been concomitantly ingested.

12. PHARMACOLOGICAL PROPERTIES

12.1 Pharmacodynamic properties

Pharmacotherapeutic group: antihistamine for systemic use, piperazine derivative
ATC code: R06A E09

Mechanism of action
Levocetirizine, the active enantiomer of cetirizine, is an antihistamine; its principal effects are mediated via selective inhibition of H1 receptors.

Pharmacodynamics
Levocetirizine, the (R) enantiomer of cetirizine, is a potent and selective antagonist of peripheral H1-receptors. Binding studies revealed that levocetirizine has high affinity for human H1-receptors. Levocetirizine dissociates from H1-receptors with a half life of 115 ± 38 min. After single administration, levocetirizine shows a receptor occupancy of 90% at 4 hours and 57% at 24 hours.

Pharmacokinetic/pharmacodynamic relationship: The action on histamine-induced skin reactions was out of phase with the plasma concentrations. ECGs did not show relevant effects of levocetirizine on QT interval.

12.2 Pharmacokinetic properties

The pharmacokinetics of levocetirizine are linear with dose- and time-independent with low inter-subject variability. The pharmacokinetic profile is the same when given as the single enantiomer or when given as cetirizine. No chiral inversion occurs during the process of absorption and elimination.

Absorption
Levocetirizine is rapidly and extensively absorbed following oral administration. Peak plasma concentrations are achieved 0.9 hour after dosing. The extent of absorption is dose-independent and is not altered by food, but the peak concentration is reduced and delayed.

Distribution
No tissue distribution data are available in humans, neither concerning the passage of levocetirizine through the blood-brain-barrier. Levocetirizine is 90% bound to plasma proteins. The distribution of levocetirizine is restrictive, as the volume of distribution is 0.4 L/kg.

Metabolism
The extent of metabolism of levocetirizine in humans is less than 14% of the dose and therefore differences resulting from genetic polymorphism or concomitant intake of hepatic drug metabolizing enzyme inhibitors are expected to be negligible. Metabolic pathways include aromatic oxidation, N- and O-dealkylation, and taurine conjugation. Dealkylation pathways are primarily mediated by CYP 3A4 while aromatic oxidation involves multiple and/or unidentified CYP isoforms. Due to its low metabolism and absence of metabolic inhibition potential, the interaction of levocetirizine with other substance, or vice-versa, is unlikely.

Elimination
The plasma half-life in adults is 7.9 ± 1.9 hours. The half-life is shorter in small children. The major route of excretion of levocetirizine and metabolites is via urine, accounting for a mean of 85.4% of the dose. Excretion via faeces accounts for only 12.9% of the dose. Levocetirizine is excreted both by glomerular filtration and active tubular secretion.

Renal impairment
The apparent body clearance of levocetirizine is correlated to the creatinine clearance. It is therefore recommended to adjust the dosing intervals of levocetirizine, based on creatinine clearance in patients with moderate and severe renal impairment. In end-stage renal disease subjects, the total body clearance is decreased by approximately 80% when compared to normal subjects.

Paediatric patients
In children 6 to 11 year of age, C max and AUC values were about 2-fold greater than in adults. Total body Cl was 30% greater and elimination half-life was 24% shorter when compared with adults.

Geriatric patients
The disposition of racemic cetirizine has been shown to be dependent on renal function rather than on age. This finding would also be applicable for levocetirizine, as levocetirizine and cetirizine are both predominantly excreted in urine. Therefore, the levocetirizine dose should be adjusted in accordance with renal function in elderly patients.

Gender
Half-life was slightly shorter in women than in men; however, the body weight–adjusted oral Cl in women appears to be comparable with that in men.

Hepatic impairment
Levocetirizine has not been studied in patients with hepatic impairment.

13. STORAGE CONDITION AND EXPIRY DATE

Store below 30 °C.
Keep out of the reach and sight of children.
Do not use after the expiry date stated on the carton box and blister.

14. SHELF LIFE

3 years from the date of manufacture.

15. NATURE AND CONTENTS OF CONTAINER

The tablets are packed in aluminium/aluminium blisters placed in cardboard boxes containing 10, 100 and 500 tablets.

16. MANUFACTURER

Standard Chem. & Pharm. Co., Ltd.
No. 6-20, Tuku, Tuku Village, Sinying District, Tainan City 73055, Taiwan.

17. PRODUCT REGISTRATION HOLDER

MD Healthcare Sdn Bhd
Suite 9D, Level 9, Menara Ansar, 65 Jalan Trus, 80000 Johor Bahru, Malaysia.

18. DATE OF REVISION

22 April 2019